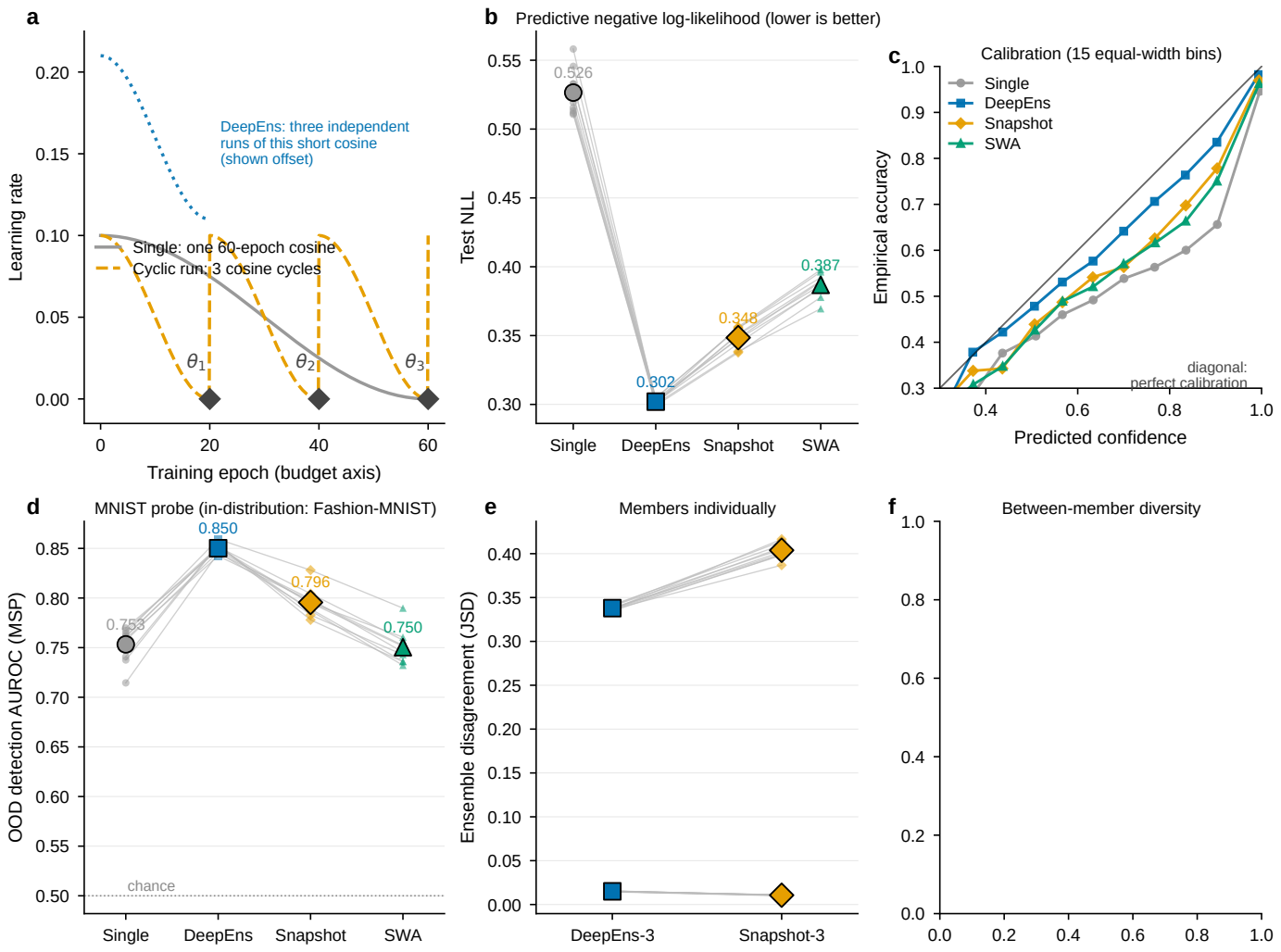
a
Learning rate
0.20
0.15
0.10
0.05
0.00
DeepEns: three independent runs of this short cosine (shown offset)
Single: one 60-epoch cosine
Cyclic run: 3 cosine cycles
θ_1
θ_2
θ_3
0
20
40
60
Training epoch (budget axis)
b
Predictive negative log-likelihood (lower is better)
Test NLL
0.55
0.50
0.45
0.40
0.35
0.30
0.526
0.302
0.348
0.387
Single
DeepEns
Snapshot
SWA
c
Calibration (15 equal-width bins)
Single
DeepEns
Snapshot
SWA
Empirical accuracy
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3
diagonal:
perfect calibration
0.4
0.6
0.8
1.0
Predicted confidence
d
MNIST probe (in-distribution: Fashion-MNIST)
OOD detection AUROC (MSP)
0.85
0.80
0.75
0.70
0.65
0.60
0.55
0.50
0.850
0.796
0.753
0.750
chance
Single
DeepEns
Snapshot
SWA
e
Members individually
Ensemble disagreement (JSD)
0.40
0.35
0.30
0.25
0.20
0.15
0.10
0.05
0.00
DeepEns-3
Snapshot-3
f
Between-member diversity
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0